

Module 6: Allocating Budget and Measuring Success

Module Takeaways

The key takeaways in this module include:

- Understand and describe how to measure the success of digital marketing campaigns.
- Explain why measuring successful marketing campaigns requires qualitative as well as quantitative analysis.
- Describe the limitations of return on investment as a metric and explain how attribution is an important consideration in measuring the success of digital marketing campaigns.
- Describe additional factors that organizations must account for when creating a digital marketing budget and explain their relevance.
- Understand customer lifetime value (LTV) and how it impacts marketing decisions.
- Differentiate between metrics that capture short-term vs. long-term marketing effectiveness and explain the key considerations in calculating each of them.
- Apply your learning in the course by allocating a digital marketing budget for an organization.
- Describe major trends that may impact the digital marketing landscape in the years ahead.

Budget Allocation

Let's examine how you can make budget allocation decisions, what factors contribute to those decisions, and what metrics are most useful to inform them. Budget allocation requires managers to have a firm grasp on which of their marketing efforts have been successful to make informed decisions. A key task in budget allocation is measuring the success of recent marketing efforts.

Customer Acquisition Cost (CAC) and Return on Investment (ROI)

Customer acquisition cost (CAC) measures how much a company spends to acquire new customers and can be calculated by dividing the total marketing spend by the number of new customers.

While CAC is a useful metric to decide where to allocate your resources, it fails to account for revenue generated. It is possible that the lowest CAC channel might be attracting customers to your low-priced products, or it might be attracting price sensitive customers to your site.

We should combine CAC with the profit generated through that channel to get the full picture of its return on investment.

Return on Investment (ROI) = (Profit – Ad Spend) / Ad Spend

Although CAC and ROI are important metrics to guide your budget allocation, several other factors should also be considered when allocating your marketing budget:

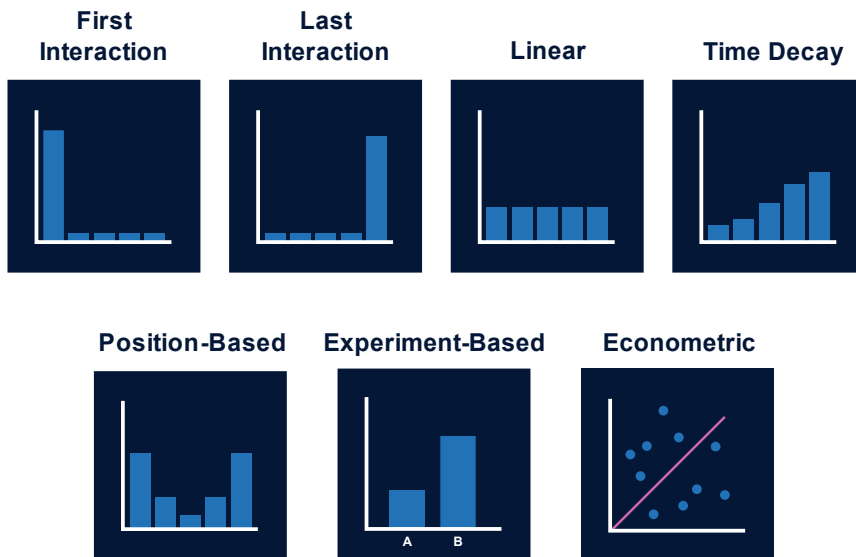
- 1) **The potential volume that can be generated from a channel.** Only a certain proportion of your target audience may be using a channel, and no matter how much you spend there, you are unlikely to increase volume after a certain point in time.
- 2) **Your desire to learn by experimentation.** A company may not have figured out the best way to utilize a channel even though it may be an appropriate channel for its target audience.
- 3) **Quality of lead.** Not all leads provide the same value to companies. Even at very low CAC, some customers are less valuable to companies—perhaps they switch brands often, or they are very price sensitive and only buy on discounts.
- 4) **The marginal cost, and not just the average cost, of customer acquisition.** As you spend more money on a channel, it gets harder and more expensive to attract customers. In other words, there are diminishing returns to scale and CAC increases as you spend more money on a channel.
- 5) **Goals.** Short-term sales might not be the only goal of a company; for example, it might also want to increase awareness.
- 6) **Attribution.** This term refers to linking marketing efforts to the customers they acquired.

Attribution

Attribution is one of the biggest challenges in budget allocation. Without understanding attribution, you may give credit and allocate more budget to lower funnel channels such as search, even though middle or upper funnel channels such as display might have helped make search more effective.

It can be difficult to evaluate the role a particular ad played in the decision a consumer makes to purchase a product. There are many methods that companies use to account for attribution. Here are a few common approaches:

- **First interaction** (also called the “first touch”) attributes 100% of the sale to the first ad the customer was exposed to.
- **Last interaction** (or the “last touch”) gives 100% credit to the last ad that a customer saw or interacted with.
- **Linear attribution** gives equal weight to all ads that a customer is exposed to.
- **Time decay attribution** gives more weight to recent ads seen by a customer.
- **Position-based attribution** takes all ads into account but weighs the first and last ads higher.
- **Experiment-based attribution** uses A/B testing to determine the role different ads play in generating revenue.
- **Econometric** models capture the effects of various ads on consumer behavior. This approach is similar to running a regression with multiple variables to find the marginal effect of each variable.



Customer Lifetime Value (LTV)

Investment in customer engagement may create value for customers, but a successful marketing strategy creates long-term value for both customers and companies.

The Two Sides of Customer Value

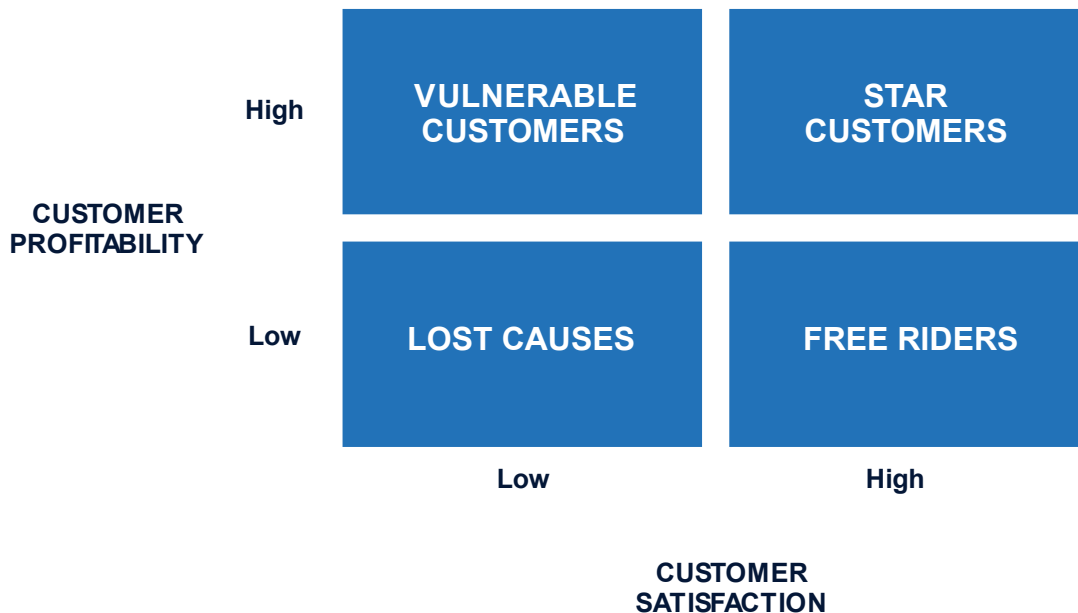
Spending money to acquire a customer is an investment. It might be beneficial to spend more on acquiring a customer than the price of a product if this customer makes multiple purchases in the future. In other words, consider the long-term profitability of customers and not only short-term sales.

Customer profitability can be combined with customer satisfaction to understand the two sides of customer value: the value that a company provides to customers and the value that it captures from customers to generate profits.

A simple way to observe this is in a 2x2 matrix Customer Value Grid.



TWO SIDES OF CUSTOMER VALUE



- **Star customers** are customers that receive high value from the firm and deliver high value to it.
- **Lost causes** are customers that derive little value from a company's product and provide little profit in return.
- **Free rider** customers take advantage of all the offers, discounts and promotions and are therefore very happy, but they are not very profitable.
- **Vulnerable** customers are highly valuable to the company but receive little value in return.

Advantages of LTV

- LTV is a useful metric in subscription-based services.
- LTV is also useful in non-contractual contexts where customers make frequent purchases over time but can stop at any time without informing the company.
- LTV is a commonly used metric by startups and venture capital firms to assess the health of a business because initially a firm loses money to acquire customers, but this is an investment that will pay off in the future as customers start buying from the firm.

Calculating LTV

Margin, churn, and customer selection are the main variables we use to calculate the LTV of various customer groups and we can use them to identify key drivers to improve LTV.

Firms can improve LTV by becoming more selective in who they acquire. They can improve churn by providing better service or retention offers, and they can improve margin by up-selling or cross-selling.

- **Customer's expected lifetime with the company can be estimated as:**

$$T = (1 / 1-r) = (1 / \text{churn rate})$$

- r = annual retention rate
- $(1-r)$ = annual churn rate

- **Lifetime value can be estimated as: $LTV = m * T$**

- m = a customer's annual contribution margin
- T = expected life (years) of a customer with the firm

Key Drivers of LTV

Acquiring and retaining customers is essential, but LTV shows us that firms must also work to develop the less-valuable customers into more valuable ones. Let's consider three ways that companies can increase the value of their existing customers.

- **Share of wallet.**
- **Cross-sell or upsell.** Once a company has a better idea of their customers' wallet shares, they can target their marketing strategies for cross-selling or upselling to these customers.
- **Redefine the scope of your business.** Organizations can foster growth by expanding into adjacent product and service categories.

Key drivers of LTV	
Customer acquisition	You can defray the cost of acquiring customers over time by strategically acquiring customers that provide more value and are less likely to churn.
Customer retention	Identifying why your customers churn allows you to target specific types of defectors with different tactics.
Customer development	You can increase the value of your existing customers by increasing your firm's share of their wallet, either by cross-selling and upselling or by adding new products and services to your business.

Time Value of Money

In calculating LTV, we cannot ignore the **time value of money**. A dollar today is more valuable than a dollar tomorrow (like an interest rate at a bank).

Because LTV captures profits over many years, it is important to incorporate the time value of money in LTV calculations.

Start with the simple LTV equation: $LTV = \text{annual margin} \times \text{the expected life of a customer}$. However, this formula does not account for the time value of money. To account for this, we can add another variable to represent discount rate:

If we assume constant margin (m), retention rate (r), and discount rate (i), this is the equation for LTV that accounts for the time value of money:

$$LTV = \frac{m}{1 - r + i} = \frac{\text{margin}}{\text{churn} + \text{discount}}$$

Relationship Between CAC, ROAS, ROI, and LTV

You're familiar with several metrics that companies can use to measure the success of marketing campaigns: customer acquisition cost (CAC), return on ad-spend (ROAS), return on investment (ROI), and now customer lifetime value (LTV). Let's examine how they are related.

Remember $ROI = (\text{Profit} - \text{Ad Spend}) / \text{Ad Spend}$

If “ n ” customers are acquired at a cost equal to CAC and long-run profit of a customer is LTV (without CAC), then:

total profit = $n * LTV$, and Ad spend = $n * CAC$

and **$ROI = [(LTV - CAC) * n] / n * CAC = (LTV / CAC) - 1$**

Adobe's Data Driven Operating Model

The Adobe simulation you completed in the course offers some options for budget allocation strategy that you may be able to apply in your own career.

Switching to a subscription model allowed Adobe to collect valuable customer data, but it lacked a consistent method for analyzing and implementing it. Each department was collecting its own data, and an understanding of the customer journey was siloed in departments. No one had a comprehensive view and departments were operating towards their own independent KPIs.

These issues led Adobe to streamline its operations around what it calls the “Data Driven Operating Model” or DDOM. Through DDOM, Adobe attempted to anchor its entire organization in the customer journey.

Fixed and Variable Investments

Fixed and variable investments affect different parts of the customer journey.

- **Fixed investments** are long-term investments that improve conversion rate at a particular stage. For example, fixed investment in landing page optimization would increase the percentage of website visitors who try the product (that is, the conversion rate from discover to try).
- **Variable investments** have a short-term effect on KPIs in that quarter. For example, increasing the amount spent on awareness marketing in a quarter would increase traffic to your website in that quarter.

Let's start with decisions about **fixed investments**. There are three questions to address: when, where, and how much to invest.

- **When should you make the fixed investments?** Because fixed investments have long-term effects, you should make these investments as soon as possible. You also need to continue investments in these instruments if you want them to have sustained impact.
- **Where, at which part of the customer journey, should you make higher fixed investments?** Your goal should be to minimize friction for consumers to buy and use your product, especially for those who are ready and inclined to buy it.
- **How much should you invest?** This is a harder question. You could also experiment with different levels of investments and monitor their impact on conversion rates.

The next task is to think about allocating **variable investments** across the customer journey.

- Directing funds towards one part of the funnel has effects on others.
- Fixed and variable investments will pay off in different ways because they solve different problems.
- Understanding the root cause of a “leak” in your funnel will help your budget allocation decisions create the greatest impact on your bottom line.
- Not all aspects of the funnel will react the same way to additional funds. Some will reach points of diminishing returns.
- Testing out different strategies and closely tracking their impact on your KPIs is the best way to maximize the impact of your budget allocation.

There is no surefire formula for success, you must test and learn to understand the ins and outs of your particular product and your particular audience.

Course Conclusion

Digital marketing is a highly dynamic and responsive industry. There will always be new trends and new developments to stay on top of. The field is constantly evolving, and strategy must evolve with it. We will always need to be ready to adapt to the next change as it comes. We encourage you to continue learning and exploring new ideas to stay ahead of the curve.

- We began this course by discussing how digital marketing and direct distribution have allowed new players to enter mature markets and take a share away from large dominant brands. This trend has created an urgency among legacy brands to learn new ways to reach and engage customers through digital marketing tools.
- We covered the fundamental building blocks of a good digital marketing plan: setting objectives, identifying target audience, articulating a clear value proposition, and having a set of KPIs or metrics to measure the success of your digital marketing efforts.
- We discussed how companies can use paid, owned, and earned media to acquire customers and how they can use personalization, story making, and brand communities to engage and retain customers.



- We elaborated on how to allocate your marketing budget and measure the effectiveness of your marketing spend using a variety of metrics such as ROAS, ROI, and LTV. Budget allocation requires a deep understanding of customer journey and a careful balance between short-term performance marketing goals and long-term brand building objectives.

We hope that you have been inspired by examples of real companies and are now equipped with the tools and frameworks to make informed digital marketing decisions for your own organization. Thank you for joining us on this digital marketing journey. We hope that this course has been a valuable learning experience for you, and we wish you all the best in your future endeavors.